

French crop yield, area and production data for ten staple crops from 1900 to 2018 on county resolution (<https://doi.org/10.5880/PIK.2021.001>)

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1. Licence

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2. Citation

When using the data please cite:

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3. Data Description

We present a data set of crop performance in France, one of Europe's major crop producers. The data set comprises ten crops (barley, maize, oats, potatoes, rapeseed, sugarbeet, sunflower, durum wheat, soft wheat and wine) and covers the years 1900 to 2018. It contains harvested area, production and yield data for all 96 French *départements* (i.e. counties or NUTS3 level) with a total number of 375,264 data points. Entries until 1988 have been digitized manually from statistical yearbooks. The technical validation indicates a high consistency of the data set within itself and with external resources. The data set may contribute to an enhanced understanding of the manifold influences on agricultural performance.

3.1. Analytical procedure:

Some yield values had to be considered as outliers, also after checking for digitizing errors. There were four criteria for defining an outlier. First, absolute yield values larger than a physiologically currently unreachable threshold were removed; threshold values were 15 t/ha for barley and durum wheat, 200 t/ha for sugarbeet and potatoes, 20 t/ha for maize, oats and wheat, 10 t/ha for rape and sunflower and 200 hl/ha for wine. Additionally, all yield values for winter rape in 1944, spring rape in 1968 and spring barley in 1980 were removed due to wrongly reported values in the yearbooks. This first step removed in total 167 yield data points. Second, the top 1% of yield values across all departments per decade were removed.

Third, values above or below the mean $\pm$ four times the standard deviation of each crop-department time series (for yield, area and production separately) were removed. Fourth, and finally, a similar variance filter as in the third step was applied within each decade of a single time series, filtering values above or below decadal mean $\pm$ two (for yield) or three (area, production) decadal standard deviations. The latter three filters removed, on average, 3.6% of the yield and 0.2% of the area or production data, respectively. There were, as a median, 43 yield outliers per department (out of 1,260 data points on average), with a range of 4 (department *Hauts de Seine*) and 255 (*Nord*) and an interquartile range of 35-50 outliers. Outliers were masked as missing values to avoid introducing a bias from any correction.

3.2. Data processing

Nationally aggregated area, production and yield data from our data set were validated with national data from 1961 to 2018 provided by the FAO⁴. Area and production data for crops with separate spring and winter data were summed on department level to test agreement with area and production data digitized for the 'total' crop.

4. File description

4.1. File inventory

Data are available for ten crops: soft wheat (spring and winter separately), durum wheat, maize, oats (spring and winter), rapeseed (spring and winter), barley (spring and winter), potatoes, sugarbeet, sunflower and wine. The split into spring and winter crops eventually results in 18 distinct crop-cultivar types. We provide two version of the full data set, one without any corrections ("RAW") and one where the filters described above have been applied ("FILTERED").

4.2. File naming convention

The file name convention is "[crop]_[season-type]_data_1900-2018_[filter-type].txt"; an example filename is "barley_winter_data_1900-2018_FILTERED.txt". Wine data only cover the years 1900-2016, but follow the same naming convention.

4.3. Description of data tables

Crop (<i>French name</i>)	Seasonal type	Years with data	Filtered outliers (<i>fraction of data</i>)			Number of data points after filtering		
			Yield	Area	Production	Yield	Area	Production
Barley (<i>Orge</i>)	Spring	1943-2018	228 (3.7%)	8 (0.1%)	10 (0.2%)	5,932	6,805	6,246
	Winter	1943-2018	243 (3.7%)	0 (0%)	4 (0.1%)	6,262	6,563	6,831
	Total	1900-2018	404 (3.7%)	9 (0.1%)	10 (0.1%)	10,381	10,783	10,784
Sugarbeet (<i>Betterave</i>)	(<i>n.a.</i>)	1900-2018	175 (3.5%)	13 (0.2%)	16 (0.3%)	4,783	5,225	5,103
Maize (<i>Maïs</i>)	(<i>n.a.</i>)	1900-2018	326 (3.7%)	3 (0.0%)	8 (0.1%)	8,452	8,793	8,784
Oats (<i>Avoine</i>)	Spring	1943-2018	232 (3.7%)	18 (0.3%)	18 (0.3%)	6,112	6,376	6,365
	Winter	1943-2018	203 (3.4%)	10 (0.2%)	14 (0.2%)	5,730	5,957	5,952
	Total	1900-2018	424 (3.9%)	1 (0.0%)	3 (0.0%)	10,341	10,781	10,777
Potatoes (<i>Pommes de terre</i>)	(<i>n.a.</i>)	1900-2018	498 (4.6%)	3 (0.0%)	52 (0.5%)	10,238	10,744	10,690
Rape (<i>Colza</i>)	Spring	1943-2018	67 (2.6%)	47 (1.6%)	14 (0.5%)	2,556	2,976	2,826
	Winter	1944-2018	165 (2.9%)	1 (0.0%)	6 (0.1%)	5,469	5,776	6,111
	Total	1900-2018	270 (3.3%)	3 (0.0%)	9 (0.1%)	7,830	8,161	8,219
Sunflower (<i>Tournesol</i>)	(<i>n.a.</i>)	1943-2018	110 (3.0%)	3 (0.1%)	5 (0.1%)	3,603	3,766	3,734
Soft wheat (<i>Froment, Blé</i>)	Spring	1943-2018	167 (3.3%)	63 (1.2%)	55 (1.1%)	4,939	5,110	5,092
	Winter	1943-2018	246 (3.5%)	4 (0.1%)	1 (0.0%)	6,759	7,009	7,010
	Total	1900-2018	378 (3.5%)	1 (0.0%)	2 (0.0%)	10,438	10,822	10,818
Durum wheat (<i>Blé dur</i>)	Total	1961-2018	92 (3.3%)	5 (0.2%)	5 (0.2%)	2,682	2,881	2,832
Wine (<i>Vignoble</i>)	(<i>n.a.</i>)	1900-2016	339 (3.9%)	7 (0.1%)	45 (0.5%)	8,435	8,816	8,804
Total yield data points			4,567 (3.6%)	199 (0.2%)	277 (0.2%)	120,942	127,344	126,978

5. References

The data set presented here has been used in two previous studies. The first describes the trends in French yields and discusses possible reasons for recently observed stagnation tendencies, while the second identifies major weather-related hazards for crop production in France. For further discussions about the crop performance data we refer to these studies.

Schauberger, B. et al. (2018). Yield trends, variability and stagnation analysis of major crops in France over more than a century. *Scientific Reports* 8, 16865, <https://doi.org/10.1038/s41598-018-35351-1>

Schauberger, B., Makowski, D., Ben-Ari, T., Boé, J. & Ciais, P. (2021). No historical evidence for increased vulnerability of French crop production to climatic hazards. *Agricultural and Forest Meteorology* 306, 108453, <https://doi.org/10.1016/j.agrformet.2021.108453>

Schauberger, B., Kato, H., Kato, T. *et al.* French crop yield, area and production data for ten staple crops from 1900 to 2018 at county resolution. *Sci Data* 9, 38 (2022). <https://doi.org/10.1038/s41597-022-01145-4>